Problem 00000002101: an arithmetic obstruction to the printed square-root/unit conjunction

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Precise scope of the refutation

The conjecture says that the Stokes constants are units of $\mathbb{Q}(\sqrt{5})$ and prints the explicit value $\varphi^{1/2}$. We use the usual golden-ratio convention

$$\varphi = \frac{1+\sqrt{5}}{2}.$$

The source supplies no other numerical definition of φ. We prove that *every* complex square root of this number lies outside the actual embedded quadratic field $\mathbb{Q}(\sqrt{5})$. Consequently no scalar can satisfy both the printed explicit-value and field-unit conditions.

This is an arithmetic obstruction to a necessary conjunction. We do not assign a value to a Stokes constant, compute any actual Stokes data, or disprove the field-unit assertion considered by itself. No assertion against general DT–RH theory or an unstated alternative meaning of φ is made. Bridgeland's actual Kronecker BPS data and RH jump conditions are given in Sections 2.9 and 3.1, printed pages 9–10 of [1]; there the Stokes factors are Poisson automorphisms of a torus. That reference provides context, not a proof or definition of the conjectured numerical value.

A genuine quadratic field and its embeddings

Construct

$$F = \mathbb{Q}[\omega], \qquad \omega^2 = 5,$$

using the native `QuadraticAlgebra` with parameters $5, 0$. There is no rational r with $r^2 = 5$: otherwise comparison with the two real roots $\pm\sqrt{5}$ would contradict the irrationality of $\sqrt{5}$. This proved fact supplies the actual native `Field` instance. The genuine basis $1, \omega$ gives $[F : \mathbb{Q}] = 2$.

The real embedding and actual conjugation are

$$\iota(a + b\omega) = a + b\sqrt{5}, \qquad \sigma(a + b\omega) = a - b\omega.$$

Conjugation is an algebra automorphism, and ι is an injective algebra homomorphism. Its range equals the native intermediate field $\mathbb{Q}(\sqrt{5}) \subset \mathbb{R}$, defined as the field adjoined by $\sqrt{5}$. Indeed the range contains $\sqrt{5} = \iota(\omega)$; conversely every $a + b\sqrt{5}$ belongs to the adjoined field. These inclusions give an actual algebra equivalence between F and that full field, not just a coordinate-vector identification. Composing with $\mathbb{R} \hookrightarrow \mathbb{C}$ gives the corresponding complex embedding; its range is likewise exactly the actual complex intermediate field adjoined by $\sqrt{5}$.

The native field norm

For $z = a + b\omega$, multiplication by z on the genuine basis has matrix

$$M_z = \begin{pmatrix} a & 5b \\ b & a \end{pmatrix}.$$

The native `Algebra.norm`, defined by this endomorphism's determinant, is therefore

$$\mathrm{N}_{F/\mathbb{Q}}(a + b\omega) = a^2 - 5b^2, \qquad z\,\sigma(z) = \mathrm{N}_{F/\mathbb{Q}}(z) \cdot 1.$$

The proof explicitly identifies the quadratic-algebra norm with this native determinant norm. It also transports the native norm through the actual real and complex field equivalences, so the calculations hold in both genuine adjoined fields.

Let $g = (1 + \omega)/2$, concretely the field element with coordinates $(1/2, 1/2)$. Its real image is φ, its conjugate's real image is $(1 - \sqrt{5})/2$, and the computed native norm is

$$\mathrm{N}_{F/\mathbb{Q}}(g) = \frac{1}{4} - 5\frac{1}{4} = -1.$$

The element g itself is a genuine field unit, since its real image is positive. The obstruction concerns its square roots.

All square roots are excluded

If $x \in F$ satisfied $x^2 = g$, multiplicativity of the native norm would give

$$\mathrm{N}_{F/\mathbb{Q}}(x)^2 = \mathrm{N}_{F/\mathbb{Q}}(x^2) = \mathrm{N}_{F/\mathbb{Q}}(g) = -1.$$

The left side is a square of a rational number and is nonnegative, a contradiction. Thus g has no square root in F.

More generally, let $f : F \hookrightarrow K$ be *any* field embedding and suppose $c^2 = f(g)$. If c belonged to the range of f, write $c = f(x)$. Injectivity gives $x^2 = g$, which was just excluded. Apply this to the actual real and complex embeddings. Every $c \in \mathbb{C}$ with $c^2 = \varphi$ therefore lies outside $\mathbb{Q}(\sqrt{5}) \subset \mathbb{C}$ and cannot be the image of a unit of that actual intermediate field.

In particular the actual principal real value $s = \sqrt{\varphi}$ is positive and satisfies $s^2 = \varphi$. The real and complex square-root equations have exactly the branches s and $-s$; the argument excludes both. It applies pointwise to every possible family of scalar data, so there is no need to construct or assign a replacement Stokes object to obtain the contradiction. Only the printed explicit-value/field-unit conjunction under the stated golden-ratio convention is refuted.

Formalization and reproducibility

The complete proof is in `lean/Main.lean`, namespace `TLMC2101`.

Its main theorem is `complete_counterexample`. Principal bridges include:

Bridge	Declaration
Actual quadratic-field dimension	`quadratic_degree`
True real adjoin-field identification	`realEmbedding_range`
Native determinant norm identity	`nativeNorm_eq_quadraticNorm`
Norm in the actual complex field	`actual_complex_field_golden_norm`
All embedding images exclude roots	`no_embedded_square_root`
Both complex branches	`complex_square_root_branches`
Actual field-unit conjunction fails	`no_unit_square_root`

The project pins Lean 4.33.0 and Mathlib commit

`db584cd6d46c92f209a44c0f1c829460d327499d`.

Run `lean -version` separately; `lean/verify.sh` then runs the root build, warning-as-error audit and official full fresh-kernel replay. The audit enumerates every owned declaration, including generated names, and runs ordinary `#print axioms` on each. Only `propext`, `Classical.choice` and `Quot.sound` are allowed foundational axioms.

References

[1] T. Bridgeland, *Riemann–Hilbert problems from Donaldson–Thomas theory*, arXiv:1611.03697v5 (24 June 2020), Sections 2.9 and 3.1, printed pp. 9–10. `https://arxiv.org/abs/1611.03697v5`.